



THE UNIVERSITY OF
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FINAL REPORT

The Art of Scientific Computation Power Laws and Finite Size Scaling in the BTW Abelian Sandpiles Model

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Contents

1	Introduction	2
2	Simulation	3
3	Power Law Fit	4
3.1	Naïve Power Law Fit	4
4	Finite Size Scaling Effect	7
5	Non-FSS Behaviour of Toppling Number	8
	Bibliography	8

Chapter 1

Introduction

1. Introduce the BTW sandpiles model as a model for self-organised criticality and $1/f$ noise [1]
2. Define criticality, self-organised criticality, power law scaling, etc
3. Mathematically describe the BTW model and toppling condition
4. Describe modelling choices for power law and truncated power law distributions
5. Justify the validity of distribution modelling using markov chain theory

Chapter 2

Simulation

- Describe the simulation procedure including implementation details (numba, burn-in)
- Show plots of basic results found
- Describe the qualitative scaling behaviour observed across different L

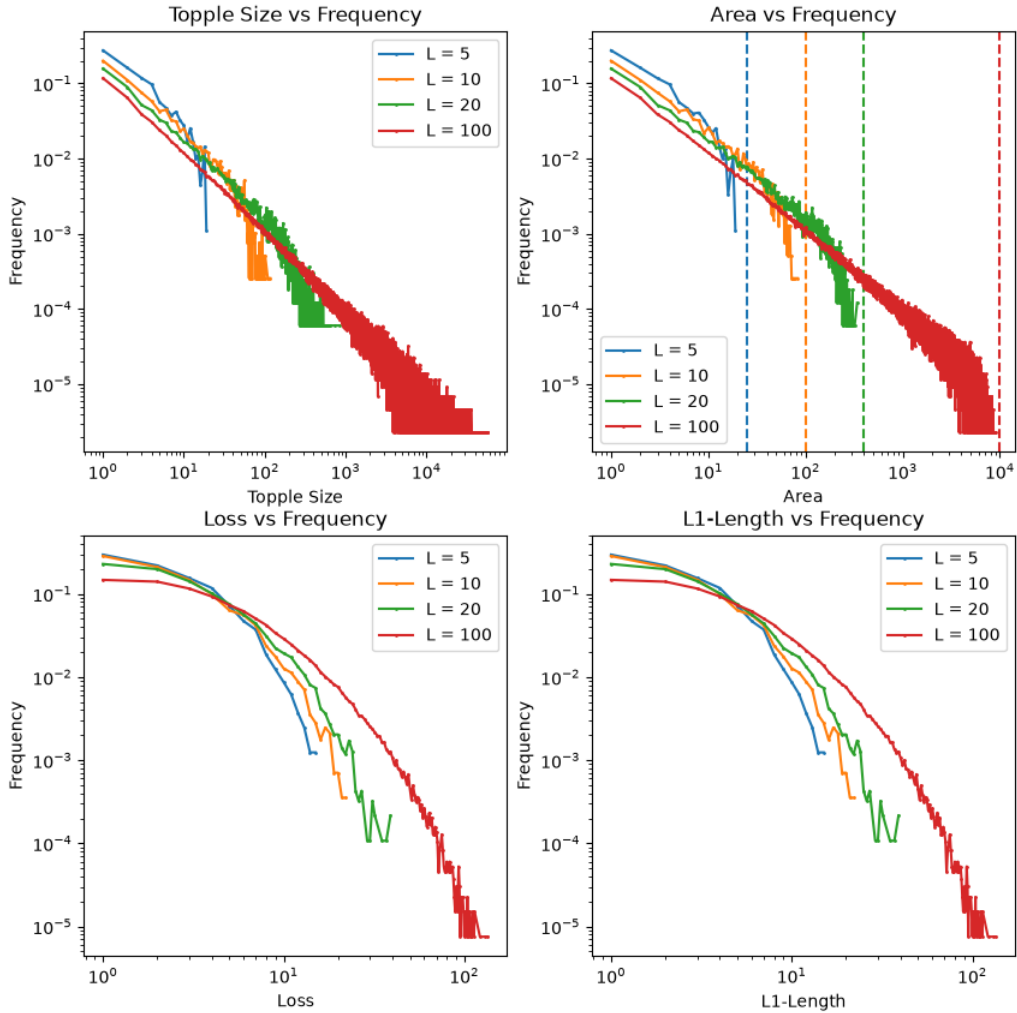


Figure 2.1: Histogram of Topple Size, Area, Loss, and L1 Length on log-log axes, for $L = 5, 10, 20, 100$. Data was generated from $100L^2$ random grain placements, with $4L^2$ burn-in. Black lines in the Area histogram (top-right) are at L^2 and represent the maximum possible toppling area obtainable from an $L \times L$ grid.

Chapter 3

Power Law Fit

1. Naive Power Law Fit
2. Hard Truncated Power Law Fit
3. (Optional) Note on Linear Regression

3.1 Naïve Power Law Fit

We hypothesize a power law scaling relationship for Toppling Size S and area A of the form:

$$p(S = s; \tau, s_{max}) \propto s^{-\tau_s} p(A = a; \tau) \propto a^{-\tau_a}$$

In the above, s_{max} and a_{max} are known constants.

Let $\mathbf{s} = (s_1, \dots, s_T)^T$, $\mathbf{a} = (a_1, \dots, a_T)^T$ be vectors representing the simulation data for topple size and area respectively.

The likelihood function for topple size (and similarly area) is:

$$\begin{aligned} \mathcal{L}(\mathbf{s}; \tau_s, s_{max}) &= \prod_{i=1}^T \frac{s_i^{-\tau_s}}{C_{\tau_s, s_{max}}} \\ \log \mathcal{L}(\mathbf{s}; \tau_s, s_{max}) &= -T \log C_{\tau_s, s_{max}} - \tau_s \sum_{i=1}^T \log s_i \end{aligned}$$

Where $C_{\tau_s, s_{max}}$ is the normalising constant:

$$C_{\tau_s, s_{max}} = \sum_{k=1}^{s_{max}} k^{-\tau_s}$$

To estimate the parameters τ_s and τ_a , we compute their Maximum Likelihood Estimates (MLEs). These are the values $\hat{\tau}_s$ and $\hat{\tau}_a$ that satisfy

$$\hat{\tau}_s = \arg \max_{\tau_s} \mathcal{L}(\mathbf{s}; \tau_s, s_{max}) = \arg \min_{\tau_s} \{-\log \mathcal{L}(\mathbf{s}; \tau_s, s_{max})\}$$

and similarly for $\hat{\tau}_a$.

Because the normalising constant $C_{\tau_s, s_{max}}$ depends on the parameter τ_s , we cannot find an analytic solution and rely on numerical optimisation techniques. The numerical optimisation technique used here is the L-BFGS-B algorithm introduced by Byrd et al. (1995) [2] and implemented in the `scipy` python library.

Initially, we ignore the non-linear tails of the distribution and model the power law as though it is valid for all plausible values. For area, there is a natural upper bound $a_{max} = L^2$. For topple size, which is in theory unbounded, we choose an arbitrary upper bound $s_{max} = 2L^2$ which exceeded all observed topple sizes to compute the normalising constant.

The estimates obtained are presented in Figure 3.1 and Table 3.1.

However, inspection of the fit against data in Figure 3.2 indicates that the estimated slope has been overestimated due to the non-powerlaw behaviour of the system at the tails. This effect appears to be most prominent in smaller lattice sizes, but still visibly impacts the fit for $L = 100$.

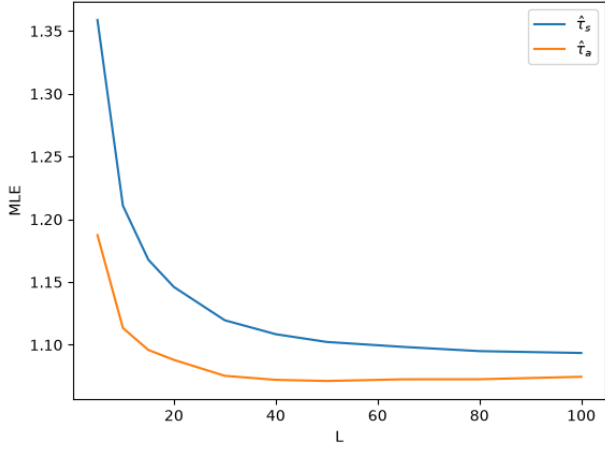


Table 3.1: MLE power-law slopes obtained for varying lattice sizes L

L	τ_s	τ_a
5	1.36	1.17
10	1.22	1.12
15	1.17	1.09
20	1.14	1.09
30	1.12	1.07
40	1.11	1.07
50	1.10	1.07
65	1.10	1.07
80	1.09	1.07
100	1.09	1.08

Figure 3.1: Estimates MLEs for τ_s, τ_a for varying lattice sizes L .

As a first correction for the finite lattice effect, we consider restricting s_{max} and a_{max} to include only those topple sizes and areas for which linearity is observed in the log-log plot. To avoid manually setting an arbitrary maximum for each value of L , here we consider the rule-of-thumb approximation $s_{max} = a_{max} = 0.1L^2$. Note in the analysis below, $L = 5$ is excluded from the study as s

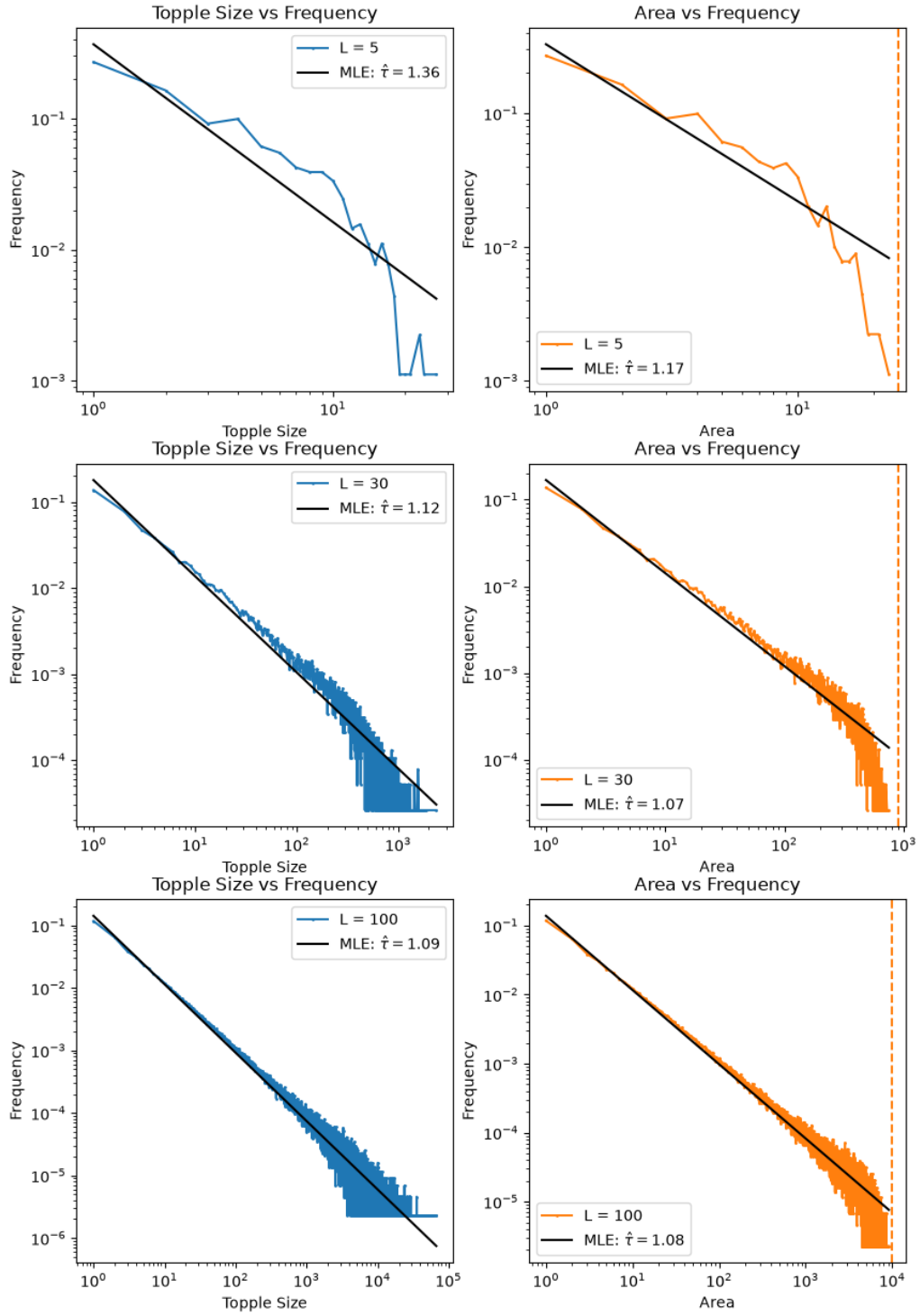


Figure 3.2: Fitted MLEs for a power law distribution against the observed data.

Chapter 4

Finite Size Scaling Effect

1. General form
2. MLE fit for exponentially-truncated power law
3. MLE fit for generalised exp-truncated power law
4. Discussion

Chapter 5

Non-FSS Behaviour of Toppling Number

1. Run-through of previous analysis with the Toppling Number
2. Identify limitations of the analysis

Bibliography

- [1] P. Bak, C. Tang, and K. Wiesenfeld. Self-organized criticality: An explanation of the $1/f$ noise. *Physical review letters*, 59(4):381, 1987.
- [2] R. H. Byrd, P. Lu, J. Nocedal, and C. Zhu. A limited memory algorithm for bound constrained optimization. *SIAM Journal on scientific computing*, 16(5):1190–1208, 1995.